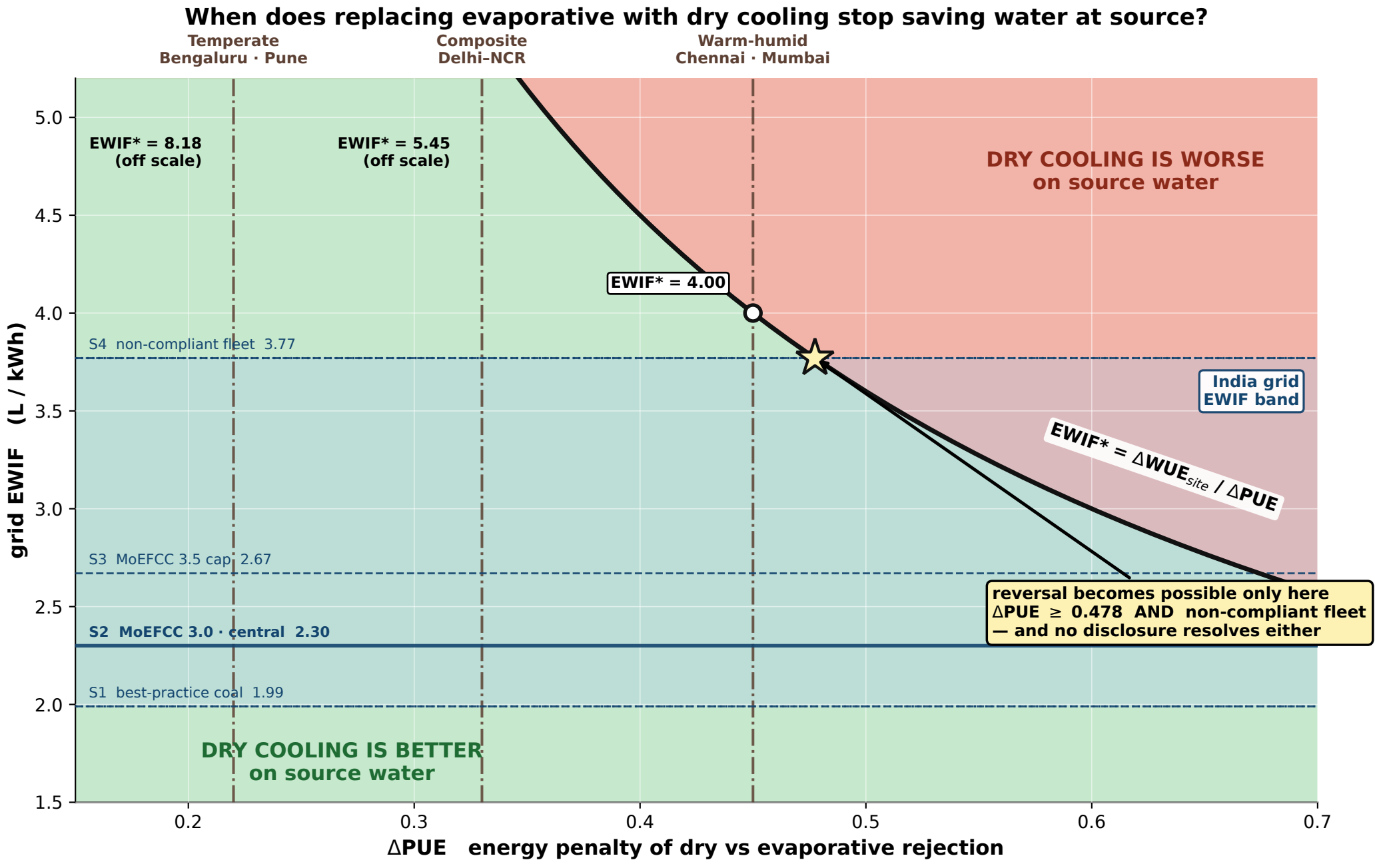
When does replacing evaporative with dry cooling stop saving water at source?
Temperate
Bengaluru · Pune
Composite
Delhi–NCR
Warm-humid
Chennai · Mumbai
EWIF* = 8.18
(off scale)
EWIF* = 5.45
(off scale)
DRY COOLING IS WORSE
on source water
EWIF* = 4.00
S4 non-compliant fleet 3.77
India grid
EWIF band
EWIF* = ΔWUE site / ΔPUE
S3 MoEFCC 3.5 cap 2.67
reversal becomes possible only here
ΔPUE ≥ 0.478 AND non-compliant fleet
— and no disclosure resolves either
S2 MoEFCC 3.0 · central 2.30
S1 best-practice coal 1.99
DRY COOLING IS BETTER
on source water
grid EWIF (L / kWh)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
0.2
0.3
0.4
0.5
0.6
0.7
ΔPUE energy penalty of dry vs evaporative rejection